A LOCAL PROPERTY OF MORPHISMS OF COMPLETE WEAKLY ADMISSIBLE RINGS STABLE UNDER BASE CHANGE IS INHERITED BY THE PROJECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one covering and one identification of a fibre product with a formal spectrum of a completed tensor product; both lie inside the counted span and are reproduced verbatim.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The equivalent conditions defining the property for a morphism of complete weakly admissible rings. Same chapter, Lemma lemma-property-defines-property-morphisms, invoked in the statement at proof lines 5231-5232. Fixes which formulation of the property has to be established for the projection.
- The definition of a formal algebraic space through coverings. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 5235-5242 and 5247-5248. Supplies the coverings used in the reduction and the statement that the pieces cover the fibre product.
- The fibre product of affine pieces is affine, countably indexed, and is the formal spectrum of the completed tensor product. Same chapter, Lemma lemma-types-fibre-products at proof lines 5243-5245 and Lemma lemma-fibre-product-affines-over-separated at proof lines 5255-5257. Identifies the affine pieces and the ring map whose property has to be shown.

Source reference. Original lemma 0AN3: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3703–3703. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ANG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5104–5104. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GBB: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4800–4800. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let P be a local property of morphisms of $WAdm^{count}$ which is stable under base change. Let $f : X \to Y$ and $g : Z \to Y$ be morphisms of locally countably indexed formal algebraic spaces over S. If f satisfies the equivalent conditions of Lemma 0ANG then so does $pr_2 : X \times_Y Z \to Z$.*

Proof. Choose a covering $\{Y_j \to Y\}$ as in Definition 0.1. For each j choose a covering $\{X_{ji} \to Y_j \times_Y X\}$ as in Definition 0.1. For each j choose a covering $\{Z_{jk} \to Y_j \times_Y Z\}$ as in Definition 0.1. Observe that $X_{ji} \times_{Y_j} Z_{jk}$ is an affine formal algebraic space which is countably indexed, see Lemma 0GBB. Then we see that

$$\{X_{ji} \times_{Y_j} Z_{jk} \to X \times_Y Z\}$$

is a covering as in Definition 0.1. Moreover, the morphisms $X_{ji} \times_{Y_j} Z_{jk} \to Z$ factor through Z_{jk}. By assumption we know that $X_{ji} \to Y_j$ corresponds to a morphism $B_j \to A_{ji}$ of WAdm^{count} having property P. The morphisms $Z_{jk} \to Y_j$ correspond to morphisms $B_j \to C_{jk}$ in WAdm^{count}. Since $X_{ji} \times_{Y_j} Z_{jk} = \mathrm{Spf}(A_{ji}\widehat{\otimes}_{B_j} C_{jk})$ by Lemma 0AN3 we see that it suffices to show that $C_{jk} \to A_{ji}\widehat{\otimes}_{B_j} C_{jk}$ has property P which is exactly what the condition that P is stable under base change guarantees. □